SIMON HALVDANSSON

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EDUCATION

NTNU Norwegian University of Science and Technology Aug 2021 - Jun 2025
Ph.D. in Mathematics
Thesis: Extensions of Quantum Harmonic Analysis and Applications to Time-Frequency Analysis
Awarded the Dimitris N. Chorafas Prize in 2026 for best PhD thesis at NTNU

Lund University Aug 2016 - Jun 2021
M.Sc. and B.Sc. in Mathematics GPA: 4.0/4.0
Master thesis: Existence and Approximations of Optimal Wavelets
Bachelor thesis: Computations with the 2D Coulomb Gas

EXPERIENCE

SINTEF Energy Research May 2025 - Present
Research Scientist *Trondheim, Norway*

- Worked on forecasting of electricity reserve markets, electricity loads and offshore production and consumption of gas
- Developed a system for LLM-grading of database events with structured outputs, ollama, and the Gemini API
- Created a system for fetching and maintaining a database of industrial carbon management projects in Europe
- Applied model predictive control (MPC) and machine learning for efficient control of integrated energy systems
- Project development, writing grant applications

NTNU Norwegian University of Science and Technology Aug 2021 - May 2025
PhD Candidate *Trondheim, Norway*

- Authored 8 research articles in the fields of time-frequency analysis, functional analysis and operator theory
- Presented my work to international audiences at 10+ conferences, workshops and seminars
- Held exercise sessions, wrote problems and solutions, mentored students and graded exams (2 courses per year)
- Refereed papers for JFAA, JMP, SampTA, JFA, JGA among others

RSA Ductor Summer 2017
Web Developer Consultant *Stockholm, Sweden*

- Developed a web application to support internal business processes using AngularJS

DeLaval Summer 2015 and 2016
Web Developer *Stockholm, Sweden*

- Contributed to design and development of mobile web application and development of Java backend

PROJECTS

- Developed several Android applications with a combined total of 200 000+ downloads and €2500+ revenue
- Maintaining the Hacker News client Harmonic with 18 000+ MAU, 1000+ GitHub stars
- Writing a blog about technical problems in machine learning and time-frequency analysis

TECHNICAL SKILLS

Languages Python, Java, JavaScript, C++, C#, MATLAB
Libraries PyTorch, NumPy, SciPy, Android, LTFAT, Matplotlib, Pyomo, SCIP, ollama